DIC06
Effective Gate Capacitance

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Follow Lab 6 (PDF pages 24–25) to compare real MOS input loading with the handout's 8/16/24/32 fF fixed-capacitance approximations.

2 Specification mapping

Handout requirement (PDF page)	Implementation	Verification and scope
Step 1: four Table 6.1 cases (24–25)	01_gate_load: NL=1–4	All four cases simulated
Compare MOS and fixed-C waveform/delay (25)	8 fF per doubled MLOAD	Differences quantified; equality not assumed

3 Circuit and design derivation

Both drivers use $W_n = 0.6\,\mu$m, $W_p = 1.8\,\mu$m and $L = 0.25\,\mu$m. Each MOS load doubles the widths to $1.2/3.6\,\mu$m. NL identical loads connect in parallel to the same driver output; MOS multiplicity represents these parallel units while their output still switches. The reference case a used a unit load, and case b cascaded loads. Both are corrected to the doubled parallel loading specified in the handout.

The estimate is $C_g \approx C_{ox}(W_n + W_p)L$, with $C_{ox} = \epsilon_{ox}/t_{ox}$. The handout's 5.4 nm oxide gives about 7.87 fF per load only for its original process. The supplied encrypted model does not expose a verified oxide thickness. The handout's prose reverses W/L labels; W denotes width and L length here.

4 Simulation setup

This teaching rewrite uses a separately supplied model at 1.0 V and 25 °C. The specified topology, loads and comparative dimensions are retained. The handout's 2.5 V process results are not accep-

tance values for this model. The chosen 1 V is a simulation condition, not a verified device rating. MOS bulk terminals are explicitly connected to the corresponding supply rails.

HSPICE S-2021.09 uses RUNLVL=6, retaining waveforms, printed data and native measurements. Figures are redrawn from actual data; CSV files are in each Lab's results directory. Successful solution does not establish process reliability. Warnings from encrypted model blocks remain suppressed; the existing diagnostic did not expose actionable further details, so the warnings are not declared harmless and complete model-limit compliance is not claimed.

Input edges are 0.1 ns with a 5 ns period; transient duration is 10 ns with 1 ps printing. Both output 10–90% transitions and input/output 50% propagation delays are measured.

5 Results and specification checks

All four cases solved and measured successfully. Delays below are in ps; MOS/C denote real loads/fixed capacitors.

NL	C (fF)	$t_{pHL,MOS}$	$t_{pHL,C}$	$t_{pLH,MOS}$	$t_{pLH,C}$
1	8	76.80	63.82	139.95	88.76
2	16	112.89	85.88	241.42	127.05
3	24	148.64	107.34	339.83	164.48
4	32	184.10	128.38	439.90	201.64

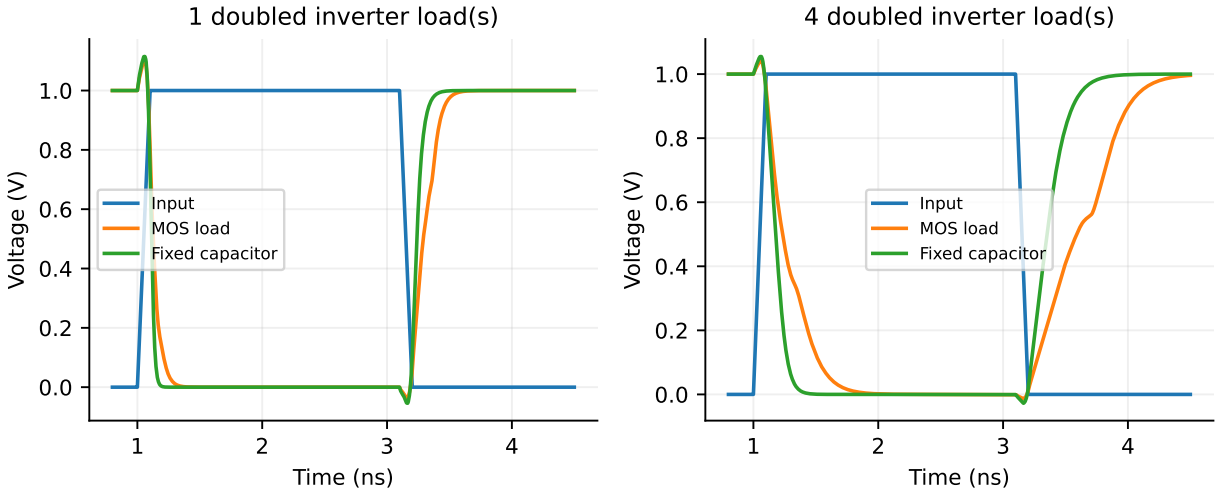


Figure 1: Waveforms with one and four loads; redrawn from this run's HSPICE data.

6 Discussion

The real MOS load is slower than the handout's fixed capacitor in all four cases, particularly for charging: NL=4 gives 439.90 ps versus 201.64 ps with 32 fF. Gate capacitance, overlap/Miller coupling and movement of the load output contribute; a single $C_{ox}WL$ value cannot capture every state. The handout's approximate equality is not reproduced under this model, a documented process-adaptation difference rather than an omitted experiment.

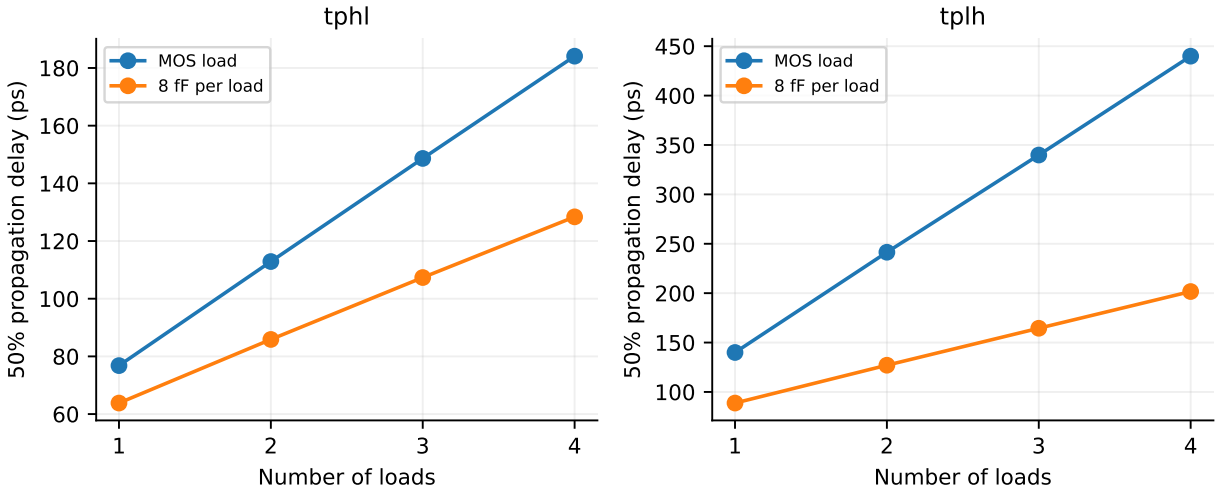


Figure 2: Propagation delay for the two load models; redrawn from this run's HSPICE data.

7 Conclusion

All four requested comparisons are complete. Eight fF per load is not a validated equivalent for this model. A subsequent cell-library model would require fitting across input slew, output loading and logic state.

8 MOS transistor schematics

Source: `01_gate_load.sp`. XDRV1 and XDRV2 use $W_n/W_p/L = 0.6/1.8/0.25\,\mu$m. The transistor load uses MNLOAD/MPLOAD with doubled widths $1.2/3.6\,\mu$m and multiplicity $M = N_L$, where $N_L = 1, 2, 3, 4$. Their drains share node z, exactly as implemented.

The drawings use the device and node names of the cited netlists. Equal net labels denote an electrical connection. Unless stated otherwise, NMOS bulk terminals connect to ground and PMOS bulk terminals connect to V_{DD}. The bulk wires are omitted from the three-terminal symbols for readability.

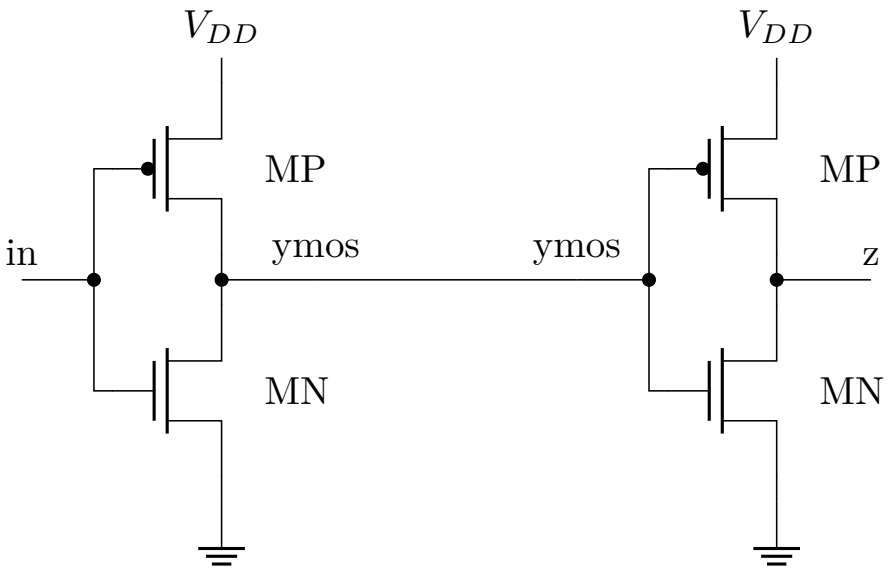


Figure 3: MOS-load branch: left cell XDRV1; right cell MNLOAD/MPLOAD with multiplicity N_L. The load output z is free to switch.

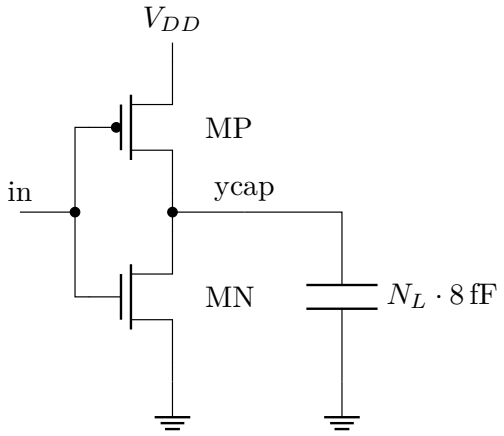


Figure 4: Capacitor comparison branch XDRV2. Both drivers receive the same input waveform.